



SCHOOL OF COMPUTER SCIENCE

BACHELOR OF COMPUTER SCIENCE (HONOURS)
 BACHELOR OF SOFTWARE ENGINEERING (HONOURS)
 BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)

ITS66904 - BIG DATA TECHNOLOGIES

SEMESTER FEBRUARY 2026

ASSESSMENT II

GROUP ASSIGNMENT (30%)

Hand-Out Date : 16th FEBRUARY 2026
Hand in Date : 12th MARCH 2026
Lecturer's Name : Liyana Rosely (Dr) & Ali Kyaw Kyaw Htike (Dr)

MLO 2:

Demonstrate practical skills in using big data tools and technologies.

Student declaration:					
1. I confirm that I am aware of the University's Regulation Governing Cheating in a University Test and Assignment and of the guidance issued by the School of Computer Science concerning plagiarism and proper academic practice, and that the assessed work now submitted is in accordance with this regulation and guidance. 2. I understand that, unless already agreed with the School of Computer Science, assessed work may not be submitted that has previously been submitted, either in whole or in part, at this or any other institution. 3. I recognise that should evidence emerge that my work fails to comply with either of the above declarations, then I may be liable to proceedings under Regulation.					
No	Student Name	Student ID	Date	Signature	Score (Office Use)
1	Rayhan Rizwan Manegar	0370148	20/02/2026		
2	Melanie Grace Yee Fone Ip Wan Fat	0373904	20/02/2026		
3	Leroy Chiew Ruo-Wey	0370618	20/02/2026		
4	Lai Ze Xuan	0371938	20/02/2026		
5	Lai Ze Qi	0371936	20/02/2026		
6	Hammad Ruxshan	0356865	20/02/2026		

APPENDIX A: GROUP INFORMATION AND CONTRIBUTIONS

TABLE 1: GROUP INFORMATION AND CONTRIBUTIONS

Group Number: *GroupBanana*

Group Leader (Name and Student ID): *Rayhan Rizwan 0370148*

No.	Student Name	Student ID	Tasks/Works Done (Specific)	Final Contribution (0/100%)	Signature
1	Rayhan	0370148	B2. Hive Implementation B5. Performance demonstration B8. Group Collaboration	16.83	
2	Melanie	0373904	A1. Problem Analysis B6. Visualisation & Interpretation B8. Group collaboration	16.66	
3	Leroy	0370618	B3. Spark SQL B5. Performance Demonstration B8. Group Collaboration	16.66	
4	Lai Ze Xuan	0371938	B4. MapReduce B5. Performance Demonstration B8. Group Collaboration	16.66	
5	Lai Ze Qi	0371936	B1. Data Description and HDFS B8. Group Collaboration	16.66	
6	Hammad	0356865	A2. Problem Analysis B7. Recommendation & Conclusion B8. Group Collaboration	16.50	

Rules: Contribution percentages must sum to 100%. Signatures may be typed or digital, but the final PDF must remain text-selectable (do not print and scan).

APPENDIX B: PEER EVALUATION MATRIX

TABLE 1: PEER EVALUATION MATRIX

Each member allocates 100 points across all members (including themselves) based on actual contribution. Compute the average for each person and use it as the Final Contribution % in Table 1.

Member's Name:

M1: *Rayhan Rizwan*

M2: *Melanie Grace*

M3: *Leroy Chiew Ruo-Wey*

M4: *Lai Ze Xuan*

M5: *Lai Ze Qi*

M6: *Hammad Ruxshan*

Evaluator / Member (1-100%)	M1	M2	M3	M4	M5	M6
M1 allocates	16	17	17	17	17	16
M2 allocates	17	17	16	17	16	17
M3 allocates	17	16	17	17	17	16
M4 allocates	17	16	16	17	17	17
M5 allocates	17	17	17	16	16	17
M6 allocates	17	17	17	16	17	16
AVERAGE	16.83	16.66	16.66	16.66	16.66	16.50

Rules: Contribution percentages must sum to 100%. Signatures may be typed or digital, but the final PDF must remain text-selectable (do not print and scan).

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A1. How big data technologies enable large-scale batch analysis for smart energy management

Smart energy management transforms continuous meter data into actionable insights which can reduce electricity costs and support sustainability. The 14-year (2004-2018) hourly demand time series (AEP_MW) dataset used in this project illustrates the difficulties associated with time-indexed energy data that grows steadily and shows the need for large-scale processing.

Large-scale analysis can be efficient by big data technologies through :

- **Managing large datasets:** Programs such as Hadoop, Hive, and Spark enable the processing of years of electricity readings to determine peak hours, seasonal patterns, and general usage trends.
- **Distributed Processing :** Compared to single-computer setups, HDFS stores data across multiple machines which allows parallel computations to be very quick.
- **Simplified aggregation :** No complicated coding is required because Hive and Spark support SQL-style queries for daily, monthly, and annual summaries.
- **Anomaly detection and prediction :** Sudden changes either rises or drops (equipment problems, energy leaks, inefficiencies) can be found in past data. Proactive solutions and precise demand forecasting can be made by machine learning that help to identify anomalies in batch or real-time (Oprea et al., 2021).
- **Operational decision support:** Managers can use analytic results to control peak demand, make the most of their schedule and reduce expenses.
- **Sustainability and efficiency :** By demonstrating where energy can be saved and how operations can impact environmental goals, energy insights support carbon-aware operations (Guato Burgos et al., 2024).

A2. Key factors for designing a scalable and efficient big data solution for smart energy analytics

Storage design and partitioning

Energy telemetry is time-indexed and typically queried over daily/weekly/monthly windows, so storage should support append-heavy ingestion and efficient time-range scans. A common Hadoop approach uses layered zones: a raw (bronze) layer to land source files for auditability, a clean (silver) layer to standardise schema/timestamps and apply data-quality rules, and a curated (gold) layer containing KPI and feature/anomaly tables for dashboards and analytics. Partitioning should be primarily time-based (e.g., year/month/day) to enable partition pruning; at the multi-meter scale, secondary clustering/bucketing by meterID/siteID improves group-by and join performance. Timestamp handling is a correctness requirement: hourly data can be affected by DST transitions and ingestion gaps/duplicates, so pipelines should normalise time consistently (e.g., UTC or a clear timezone policy) and validate expected hourly completeness, otherwise KPIs/features (e.g., "same hour last week") can be distorted.

Efficiency in the processing and choice of tools (HDFS, Hive, Spark SQL, MapReduce).

Processing efficiency and tool choice (HDFS, Hive, Spark SQL, MapReduce). Efficiency comes from matching tools to pipeline stages. HDFS provides distributed storage and parallel reads. Hive supports reproducible SQL transformations and table management for KPI aggregation/reporting. Spark SQL suits iterative feature engineering (lags, rolling windows, seasonal baselines) due to optimisation and reuse of cached intermediates (Armbrust et al., 2015). Java MapReduce remains useful for at least one scalable aggregation (e.g., daily maxima/totals) and for demonstrating distributed batch fundamentals. Across tools, mitigate the small-files problem via compaction/merging, use columnar formats (Parquet/ORC) for curated tables, push filters early (predicate pushdown where applicable), and choose appropriate join strategies for enrichment (weather/holidays/carbon intensity).

Fault Tolerance and Robustness

Failures and stragglers are expected. Robustness relies on replication, retries, and speculative execution; however, correctness also matters-missing partitions, duplicates, or partial outputs can create false anomalies, so validation rules, idempotent jobs, and checkpoints are required (Saadoon et al., 2022).

Reproducibility and auditability

Version-control code/SQL, enforce schema contracts, document parameters, log run metadata, and maintain raw → clean → curated lineage to reproduce outputs (Wang et al., 2023).

Security and privacy

Apply RBAC/least privilege, encryption, pseudonymisation, audit logs, and an aggregation-first sharing policy to protect sensitive behaviour patterns (Lee & Hess, 2021).

B1. Dataset description and HDFS ingestion

B1.1 Dataset Description

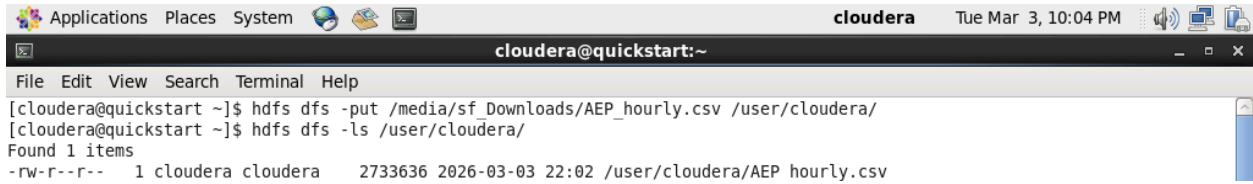
Our dataset(AEP_hourly.csv) was sourced from the Regional Transmission Organization(PJM Interconnection LLC) and was published on the Kaggle website as part of the “Hourly Energy Consumption” data collection by Rob Mulla. This particular dataset is sourced from the American Electric Power (AEP) company, which serves parts of the Midwest and Appalachian United States. The file is 2.6 - 2.7MB in size(2,670 KB) and contains 121,273 rows of data(excluding the header row) spanning from as early as December 31, 2004 to January 2, 2018. This dataset is quite minimal, containing only 2 fields. The first field is Datetime, which is a string that contains the date and time in M/D/YYYY H:MM format. The second field is AEP_MW, which is an integer that stores the demand for electricity in the AEP service territory in megawatts(MW). In terms of preprocessing steps to be performed, there are no additional columns and no null or missing values across any of the records, which means that the dataset is immediately available and useful for analysis, with no input and cleaning steps needing to be conducted.

The AEP_MW column has a wide range of values from as low as 9,581 MW to as high as 25,695 MW, with an average of about 15,500 MW. By observing the dataset using human sight, we can deduce that during nighttime the demand for electricity is lower and that peaks normally occur during summer(air conditioning) and winter(heating). Common patterns that can be observed include daytime vs nighttime, weekday vs weekends, and summer and winter values vs the other seasons.

Table B1.1: Schema Mapping table based on raw columns in AEP_hourly.csv dataset

Raw Column	Raw Type	Normalized Column	Hive Type	Spark Type
Datetime	STRING	datetime	STRING	StringType
AEP_MW	INT	aep_mw	DOUBLE	DoubleType

B1.2 HDFS Ingestion



```

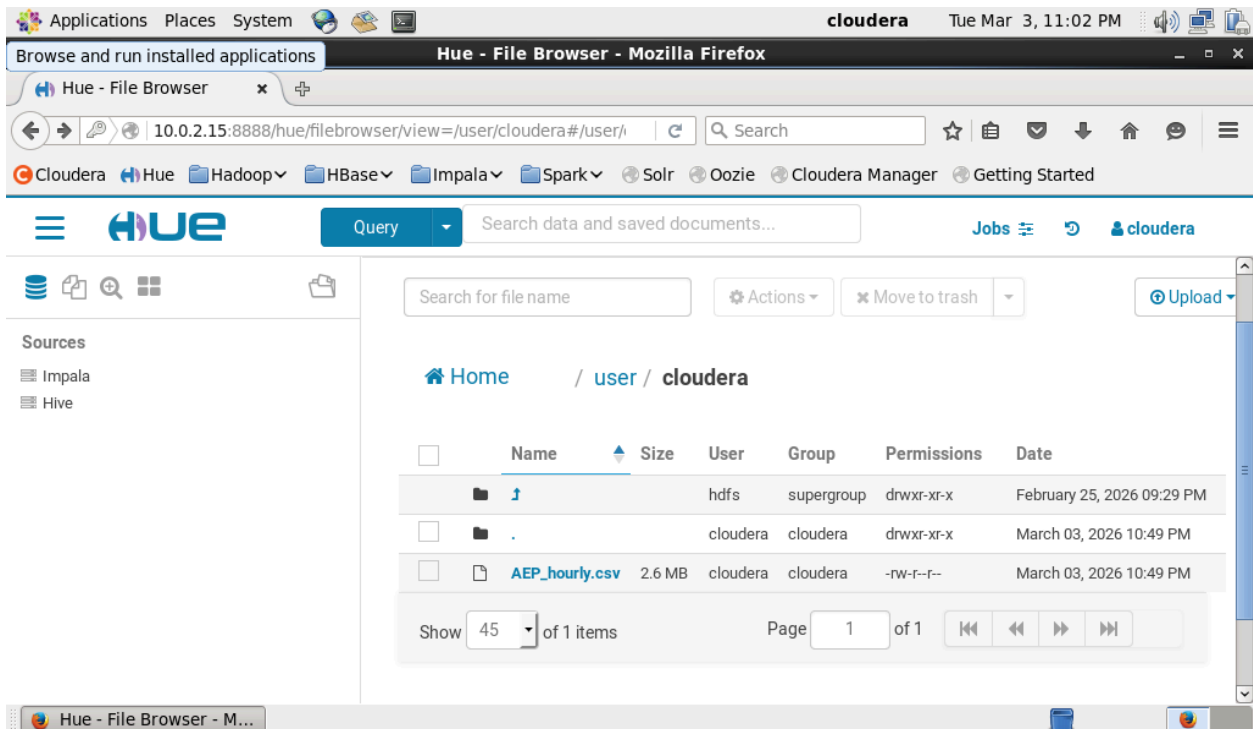
Applications  Places  System  cloudera  Tue Mar 3, 10:04 PM
cloudera@quickstart:~
File Edit View Search Terminal Help
[cloudera@quickstart ~]$ hdfs dfs -put /media/sf_Downloads/AEP_hourly.csv /user/cloudera/
[cloudera@quickstart ~]$ hdfs dfs -ls /user/cloudera/
Found 1 items
-rw-r--r--  1 cloudera cloudera    2733636 2026-03-03 22:02 /user/cloudera/AEP_hourly.csv
  
```

Figure B1.2: Evidence of uploading to HDFS

```

[cloudera@quickstart ~]$ hdfs dfs -du -h /user/cloudera/AEP_hourly.csv
2.6 M  2.6 M  /user/cloudera/AEP_hourly.csv
[cloudera@quickstart ~]$
  
```

Figure B1.3: Evidence of file size (2.6 MB)



The screenshot shows the Hue File Browser interface in Mozilla Firefox. The browser displays the file structure under the user 'cloudera'. A table lists the files and folders in the directory.

Name	Size	User	Group	Permissions	Date
↑		hdfs	supergroup	drwxr-xr-x	February 25, 2026 09:29 PM
.		cloudera	cloudera	drwxr-xr-x	March 03, 2026 10:49 PM
AEP_hourly.csv	2.6 MB	cloudera	cloudera	-rw-r--r--	March 03, 2026 10:49 PM

The interface also includes a search bar, navigation buttons, and a sidebar with sources like Impala and Hive.

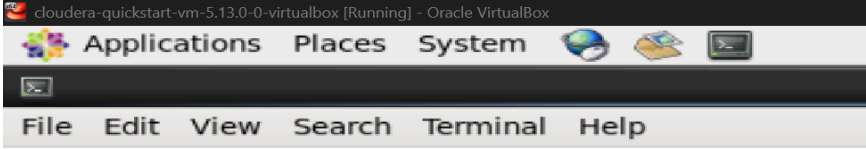
Figure B1.4: Evidence of uploading to HDFS(shown in Hue)

B2. Hive implementation

B2.1 Setting Up Hive and Loading Dataset

Apache Hive provides SQL-like querying on data stored in HDFS, making it ideal for batch analytics on large smart-energy datasets. For this implementation, an external Hive table was created to reference the HDFS dataset without data movement, enabling efficient ad-hoc queries for energy insights.

For the Hive portion of the batch analytics pipeline, the AEP hourly energy dataset was uploaded to HDFS at /user/cloudera/AEP_hourly.csv. A Hive database '**energydb**' was created, followed by an external table aep_hourly with columns datetime (STRING) and aep_mw (DOUBLE).



```

cloudera-quickstart-vm-5.13.0-0-virtualbox [Running] - Oracle VirtualBox
Applications Places System
File Edit View Search Terminal Help
[cloudera@quickstart ~]$ sudo hive

Logging initialized using configuration in file:
WARNING: Hive CLI is deprecated and migration to
hive>
>
> show databases;
OK
default
Time taken: 2.488 seconds, Fetched: 1 row(s)
hive> create database energydb;
OK
Time taken: 3.756 seconds
hive> use energydb;
OK
Time taken: 0.12 seconds
hive> show databases;
OK
default
energydb
Time taken: 0.039 seconds, Fetched: 2 row(s)
hive> █

```

Figure B2.1 Creating Database 'energydb' in Hive

```
hive> CREATE EXTERNAL TABLE aep_hourly(
> datetime STRING,
> aep_mw DOUBLE
> )
> ROW FORMAT DELIMITED FIELDS TERMINATED BY ','
```

Figure B2.2 Creating Table 'aep_hourly' in energydb

After defining the aep_hourly table structure, I loaded the hourly energy dataset from HDFS using `LOAD DATA INPATH '/user/cloudera/AEP_hourly.csv' INTO TABLE aep_hourly;`, following the data loading pattern from Practical 6. The table now contains **121274** rows from the AEP energy dataset.

```
hive> LOAD DATA INPATH '/user/cloudera/AEP_hourly.csv' INTO TABLE aep_hourly;
Loading data to table energydb.aep_hourly
Table energydb.aep_hourly stats: [numFiles=1, totalSize=2859900]
OK
Time taken: 1.109 seconds
hive> select * from aep_hourly limit 5;
OK
Datetime      NULL
31/12/2004 1:00 13478.0
31/12/2004 2:00 12865.0
31/12/2004 3:00 12577.0
31/12/2004 4:00 12517.0
Time taken: 0.089 seconds, Fetched: 5 row(s)
```

Figure B2.3 Loading Dataset into aep_hourly Table

```
hive> select count (*) from aep_hourly;
Query ID = root_20260304025757_86b66599-1621-4b
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile ti
In order to change the average load for a reduc
set hive.exec.reducers.bytes.per.reducer=<num
In order to limit the maximum number of reducer
set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
set mapreduce.job.reduces=<number>
Starting Job = job_1772620543003_0001, Tracking
Kill Command = /usr/lib/hadoop/bin/hadoop job
Hadoop job information for Stage-1: number of m
2026-03-04 02:57:29,863 Stage-1 map = 0%, redu
2026-03-04 02:57:44,995 Stage-1 map = 100%, re
2026-03-04 02:58:02,800 Stage-1 map = 100%, re
MapReduce Total cumulative CPU time: 2 seconds
Ended Job = job_1772620543003_0001
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative C
Total MapReduce CPU Time Spent: 2 seconds 310 m
OK
121274
Time taken: 60.138 seconds, Fetched: 1 row(s)
```

Figure B2.4 Evidence of table containing 121274 rows

To enable daily aggregation, I created `aep_hourly_parsed` to extract a proper DATE column from the datetime string using `unix_timestamp` and `to_date` [Figure B2.5]. The core analytical query groups hourly records by date and computes daily total load (`SUM(aep_mw)`) and daily peak load (`MAX(aep_mw)`), storing results in `aep_daily` [Figure B2.7].

```
hive> CREATE TABLE aep_hourly_parsed AS
> SELECT
>   from_unixtime(unix_timestamp(datetime, 'd/M/yyyy H:mm')) AS ts,
>   to_date(from_unixtime(unix_timestamp(datetime, 'd/M/yyyy H:mm'))) AS d,
>   aep_mw
> FROM aep_hourly;
Query ID = root_20260304030909_edcb99ab-62b8-4e2d-bf1d-8baeb7347a52
Total jobs = 3
```

Figure B2.5 Hive Parsed Table Creation

```
hive> select * from aep_hourly_parsed limit 5;
OK
NULL      NULL      NULL
2004-12-31 01:00:00      2004-12-31      13478.0
2004-12-31 02:00:00      2004-12-31      12865.0
2004-12-31 03:00:00      2004-12-31      12577.0
2004-12-31 04:00:00      2004-12-31      12517.0
Time taken: 0.088 seconds, Fetched: 5 row(s)
hive> █
```

```
hive> Describe aep_hourly_parsed;
OK
ts              string
d              string
aep_mw         double
Time taken: 1.054 seconds, Fetched: 3 row(s)
hive> █
```

Figure B2.6 Parsed Table Data Verification

```
hive> create table aep_daily as select
>   d,
>   sum(aep_mw) as daily_total_mw,
>   max(aep_mw) as daily_max_mw
> from aep_hourly_parsed
> group by d
> order by d;
```

Figure B2.7 Daily Energy Aggregation Query

B2.2 Queries Section

To extract meaningful energy insights from the dataset, we executed SQL-style queries using Apache Hive. The following sample query demonstrates the daily aggregation of electricity consumption, calculating both the total and peak megawatt (MW) usage per day.

Query 1: Daily Total and Peak Load

```
hive> SELECT d,
> SUM(aep_mw) as daily_total_mw,
> MAX(aep_mw) as daily_peak_mw
> FROM aep_hourly_parsed
> GROUP BY d
> ORDER BY d DESC
> LIMIT 10;
```



```
OK
2018-08-03      14809.0 14809.0
2018-08-02      376504.0      18869.0
2018-08-01      363628.0      17204.0
2018-07-31      364327.0      16984.0
2018-07-30      368834.0      18262.0
2018-07-29      327792.0      16645.0
2018-07-28      338714.0      16725.0
2018-07-27      379599.0      18641.0
2018-07-26      395382.0      20356.0
2018-07-25      396080.0      19756.0
Time taken: 189.217 seconds, Fetched: 10 row(s)
```

Figure B2.8: HiveQL query execution and output calculating daily total and peak energy load.

The query successfully groups the hourly data by date to output the daily total consumption and the maximum load recorded for that day. For example, on August 2, 2018, the total energy load was 376,504.0 MW, with a peak hourly load reaching 18,869.0 MW. Additional HiveQL queries analyzing 24-hour peak patterns and weekday versus weekend consumption behaviors are provided in the Appendix.

B3. Spark SQL implementation

Query 1: Daily Total and Peak Load

Insight:

How much electricity is used each day, and what is the highest load reached on that day

```
scala> sqlContext.sql("""
  | SELECT
  |   d,
  |   SUM(aep_mw) AS daily_total_mw,
  |   MAX(aep_mw) AS daily_peak_mw
  | FROM aep_hourly_parsed
  | GROUP BY d
  | ORDER BY d DESC
  | LIMIT 10
  | """).show()
```

d	daily_total_mw	daily_peak_mw
2018-08-03	14809.0	14809.0
2018-08-02	376504.0	18869.0
2018-08-01	363628.0	17204.0
2018-07-31	364327.0	16984.0
2018-07-30	368834.0	18262.0
2018-07-29	327792.0	16645.0
2018-07-28	338714.0	16725.0
2018-07-27	379599.0	18641.0
2018-07-26	395382.0	20356.0
2018-07-25	396080.0	19756.0

Time taken: 117.40 seconds

Rows fetched: 10

Figure B3.1: Daily Total and Peak Load

What the result shows:

The output lists the most recent 10 days in the dataset, together with their total daily energy usage and peak load. For example, some days, such as 2018-07-25 and 2018-07-26, show very high total daily consumption, while 2018-08-03 is much lower because the day appears incomplete in the dataset.

B4. Hadoop MapReduce (Java) implementation

B4.1 Code for Aggregation (Daily total energy and Peak load)

MapReduce uses Java implementation to process data. For this implementation, Java code is run to map and reduce the HDFS dataset into daily total energy and Peak load.

Mapping

```
public static class EnergyMapper
    extends Mapper<LongWritable, Text, Text, DoubleWritable> {
    private Text dateKey = new Text();
    private DoubleWritable mwValue = new DoubleWritable();

    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString().trim();
        // Skip header row
        if (line.startsWith("Datetime") || line.isEmpty()) {return;}
        // Format: "DD/MM/YYYY H:MM,MW"
        String[] parts = line.split(",");
        if (parts.length < 2) return;
        String datetime = parts[0].trim(); // e.g. "31/12/2004 1:00"
        String mwStr = parts[1].trim(); // e.g. "13478"
        // Extract date part only (before the space)
        String[] dtParts = datetime.split(" ");
        if (dtParts.length < 1) return;
        // Split DD/MM/YYYY
        String[] dateParts = dtParts[0].split("/");
        if (dateParts.length != 3) return;
        String day = dateParts[0];
        String month = dateParts[1];
        String year = dateParts[2];
        // Reformat to YYYY-MM-DD for proper sorting
        String formattedDate = String.format("%s-%02d-%02d", year, Integer.parseInt(month), Integer.parseInt(day));
        try {
            double mw = Double.parseDouble(mwStr);
            dateKey.set(formattedDate);
            mwValue.set(mw);
            context.write(dateKey, mwValue);
        } catch (NumberFormatException e) { // skip rows with non-numeric MW values
        }
    }
}
```

Figure B4.1: Mapper Code Snippet

The Mapper reads each line of the CSV file one at a time and skips the header row. It then parses the datetime string from the raw DD/MM/YYYY H:MM format and reformats it into YYYY-MM-DD for correct chronological sorting. For each valid row, the Mapper emits the formatted date as the key and the energy/Megawatt (MW) value as the value, which gets passed to the Reducer.

Reducing

```
public static class EnergyReducer
    extends Reducer<Text, DoubleWritable, Text, Text> {
    private DoubleWritable result = new DoubleWritable();

    public void reduce(Text key, Iterable<DoubleWritable> values, Context context)
        throws IOException, InterruptedException {

        double dailyTotal = 0.0;
        double dailyMax = 0.0;

        for (DoubleWritable val : values) {
            dailyTotal += val.get();
            if (val.get() > dailyMax) {
                dailyMax = val.get();
            }
        }

        // Write total
        context.write(key, new Text(dailyTotal + "\t" + dailyMax));
    }
}
```

Figure B4.2: Reducer Code Snippet

The Reducer receives all energy (in megawatt) values grouped under the same date key and loops through them to compute the daily total energy consumption by summing all hourly MW readings, and the daily peak load by tracking the highest value seen. It then writes both results as a single output line keyed by date, producing one output record per day in the dataset.

B4.2 Commands to Run

```
[cloudera@quickstart ~]$ cd GroupAssignment
[cloudera@quickstart GroupAssignment]$ nano DailyEnergyAggregation.java
[cloudera@quickstart GroupAssignment]$ HCP=$(hadoop classpath)
[cloudera@quickstart GroupAssignment]$ echo $HCP
/etc/hadoop/conf:/usr/lib/hadoop/lib/*:/usr/lib/hadoop/./*/:/usr/lib/hadoop-hdfs
/./:/usr/lib/hadoop-hdfs/lib/*:/usr/lib/hadoop-hdfs/./*/:/usr/lib/hadoop-yarn/li
b/*:/usr/lib/hadoop-yarn/./*/:/usr/lib/hadoop-mapreduce/lib/*:/usr/lib/hadoop-ma
preduce/./*/
[cloudera@quickstart GroupAssignment]$ javac -classpath "$HCP" -d . DailyEnergyAggregation.java
[cloudera@quickstart GroupAssignment]$ jar -cvf DailyEnergyAggregation.jar DailyEnergyAggregation*.class
added manifest
adding: DailyEnergyAggregation.class(in = 1614) (out= 831)(deflated 48%)
adding: DailyEnergyAggregation$EnergyMapper.class(in = 2651) (out= 1175)(deflated 55%)
adding: DailyEnergyAggregation$EnergyReducer.class(in = 1799) (out= 751)(deflated 58%)
```

Figure B4.3: Commands

DailyEnergyAggregation.java is created inside the GroupAssignment directory, with the MapReduce code entered. The hadoop classpath is stored inside a variable (HCP), and the java file is then compiled using javac with the Hadoop classpath (HCP variable) as dependencies. The absence of any error messages confirms that the Mapper, Reducer, and main class are syntactically correct and ready to be packaged into a JAR file.

```
[cloudera@quickstart GroupAssignment]$ jar -cvf DailyEnergyAggregation.jar DailyEnergyAggregation*.class
added manifest
adding: DailyEnergyAggregation.class(in = 1614) (out= 831)(deflated 48%)
adding: DailyEnergyAggregation$EnergyMapper.class(in = 2651) (out= 1175)(deflated 55%)
adding: DailyEnergyAggregation$EnergyReducer.class(in = 1799) (out= 751)(deflated 58%)
```

Figure B4.4: Creating jar file command

After the dependencies have been declared, all three compiled .class files: the main class, Mapper, and Reducer class will be bundled into a single executable jar file called DailyEnergyAggregation.jar

```
[cloudera@quickstart GroupAssignment]$ hadoop jar /home/cloudera/GroupAssignment/DailyEnergyAggregation.jar DailyEnergyAggregation /user/cloudera/AEP_hourly.csv /user/cloudera/output/daily_energy
```

Figure B4.5: Command to run Jar File

```
26/03/07 19:28:34 INFO mapreduce.Job: Running job: job_1772895249555_0003
26/03/07 19:28:38 INFO mapreduce.Job: Job job_1772895249555_0003 running in uber mode : false
26/03/07 19:28:38 INFO mapreduce.Job: map 0% reduce 0%
26/03/07 19:28:43 INFO mapreduce.Job: map 100% reduce 0%
26/03/07 19:28:49 INFO mapreduce.Job: map 100% reduce 100%
26/03/07 19:28:50 INFO mapreduce.Job: Job job_1772895249555_0003 completed successfully
```

Figure B4.6: MapReduce job execution showing Map and Reduce progress

```

Map-Reduce Framework
  Map input records=121274
  Map output records=121273
  Map output bytes=2304187
  Map output materialized bytes=2546739
  Input split bytes=125
  Combine input records=0
  Combine output records=0
  Reduce input groups=5055
  Reduce shuffle bytes=2546739
  Reduce input records=121273
  Reduce output records=5055

```

Figure B4.7: Showing number of processed and produced records during execution

The `hadoop jar` command sends the jar file to the Hadoop cluster, which splits the input CSV across Mappers and then passes the intermediate results to the Reducer, where the jar file is run using `AEP_hourly.csv` in the `/user/cloudera` directory in HDFS. The output will be stored in the `/user/cloudera/output/daily_energy` directory in HDFS. The progress lines confirm the job completed successfully with all 121,273 records processed and 5,055 daily totals produced.

```

[cloudera@quickstart GroupAssignment]$ hdfs dfs -ls /user/cloudera/output/daily_energy
Found 2 items
-rw-r--r--  1 cloudera cloudera          0 2026-03-07 19:28 /user/cloudera/output/daily_energy/SUCCESS
-rw-r--r--  1 cloudera cloudera    101099 2026-03-07 19:28 /user/cloudera/output/daily_energy/part-r-00000

```

Figure B4.8: Listing output directory in HDFS

After execution, the `/user/cloudera/output/daily_energy` directory is automatically created, and inside there are 2 files. If one says `SUCCESS`, then it confirms that MapReduce has successfully processed and written the daily energy totals and peak load into HDFS.

```

[cloudera@quickstart GroupAssignment]$ hdfs dfs -cat /user/cloudera/output/daily_energy/part-r-00000 | tail -10
2018-07-25      396080.0      19756.0
2018-07-26      395382.0      20356.0
2018-07-27      379599.0      18641.0
2018-07-28      338714.0      16725.0
2018-07-29      327792.0      16645.0
2018-07-30      368834.0      18262.0
2018-07-31      364327.0      16984.0
2018-08-01      363628.0      17204.0
2018-08-02      376504.0      18869.0
2018-08-03      14809.0 14809.0

```

Figure B4.9: Output

Finally, run the `part-r-00000` file that contains the output and print what is inside using `cat`. The `"| tail -10"` limits the table to print the last 10 rows (most recent rows), so the terminal does not overflow. Each line in the output represents one day's total electricity consumption in megawatts, and the highest energy value on that day.

B5. Performance demonstration and comparison

To evaluate the processing performance of our big data tools, we selected the daily total usage + peak load aggregation task and executed it multiple (5) times.

B5.1 Hive Implementation

The table below records the execution times for the Hive implementation over five separate runs.

```
hive> SELECT d,
> SUM(aep_mw) as daily_total_mw,
> MAX(aep_mw) as daily_peak_mw
> FROM aep_hourly_parsed
> GROUP BY d
> ORDER BY d DESC
> LIMIT 10;
```

Time taken: 152.399 seconds, Fetched: 10 row(s)

Time taken: 133.806 seconds, Fetched: 10 row(s)

Time taken: 186.402 seconds, Fetched: 10 row(s)

Time taken: 219.142 seconds, Fetched: 10 row(s)

Time taken: 128.403 seconds, Fetched: 10 row(s)

```
OK
2018-08-03      14809.0 14809.0
2018-08-02      376504.0      18869.0
2018-08-01      363628.0      17204.0
2018-07-31      364327.0      16984.0
2018-07-30      368834.0      18262.0
2018-07-29      327792.0      16645.0
2018-07-28      338714.0      16725.0
2018-07-27      379599.0      18641.0
2018-07-26      395382.0      20356.0
2018-07-25      396080.0      19756.0
```

Figure B5.1: 10 Lines of output from Hive Implementation & Time taken.

Table B5.1: Execution times for HiveQL daily aggregation task across 5 runs.

Run	Time (seconds)
1	152.933 seconds
2	133.806 seconds
3	186.402 seconds
4	219.142 seconds
5	128.403 seconds
Average	164.137 seconds

B5.2 Spark SQL

The Spark SQL query was executed five times to measure performance consistency. The first run took longer (91.38 seconds) due to initial data loading and Spark execution setup. Subsequent runs were much quicker as Spark reused cached data and optimised execution. This demonstrates how Spark improves performance after the initial execution through in-memory processing and query optimisation.

```
scala> val result = sqlContext.sql("""
  | SELECT
  | d,
  | SUM(aep_mw) AS daily_total_mw,
  | MAX(aep_mw) AS daily_peak_mw
  | FROM aep_hourly_parsed
  | GROUP BY d
  | ORDER BY d DESC
  | LIMIT 10
  | """).show()
```

d	daily_total_mw	daily_peak_mw
2018-08-03	14809.0	14809.0
2018-08-02	376504.0	18869.0
2018-08-01	363628.0	17204.0
2018-07-31	364327.0	16984.0
2018-07-30	368834.0	18262.0
2018-07-29	327792.0	16645.0
2018-07-28	338714.0	16725.0
2018-07-27	379599.0	18641.0
2018-07-26	395382.0	20356.0
2018-07-25	396080.0	19756.0

Time taken: 91.38 seconds
 Time taken: 25.63 seconds
 Time taken: 21.42 seconds
 Time taken: 18.14 seconds
 Time taken: 12.10 seconds

Figure B5.2: Output from Spark SQL & Time taken.

Table B5.2: Execution time of daily aggregation SPARK SQL query across five runs

Run	Time (seconds)
1	91.38 seconds
2	25.63 seconds
3	21.42 seconds
4	18.14 seconds
5	12.10 seconds
Average	33.73 seconds

B5.3 Hadoop MapReduce

The table below records the execution times for the MapReduce implementation over five separate runs.

```

2018-07-25      396080.0      19756.0
2018-07-26      395382.0      20356.0
2018-07-27      379599.0      18641.0
2018-07-28      338714.0      16725.0
2018-07-29      327792.0      16645.0
2018-07-30      368834.0      18262.0
2018-07-31      364327.0      16984.0
2018-08-01      363628.0      17204.0
2018-08-02      376504.0      18869.0
2018-08-03      14809.0 14809.0

```

Figure B5.3: 10 lines of Output from MapReduce Job

```

|real    0m15.477s |real    0m15.172s
|real    0m14.163s |real    0m17.131s |real    0m19.864s

```

Table B5.3: Execution times for MapReduce daily aggregation task across 5 runs.

Run	Time (seconds)
1	15.477 seconds
2	15.172 seconds
3	14.163 seconds
4	17.131 seconds
5	19.864 seconds
Average	16.361 seconds

B5.4 Comparison

Table B5.4: Average Execution time across all tools

<i>Tool</i>	<i>Average Time (seconds)</i>
Hive	164.137 s
Spark	33.73 s
MapReduce	16.361 s

The 3 tools all present different results, some slower and some faster than the rest. This is due to the difference in terms of their execution models, where for Hive, it has to translate HiveQL into MapReduce jobs internally, adding extra layers of query parsing and processing before any computation. MapReduce achieved the fastest average time despite being the lowest level tool, as it is able to execute a direct aggregation process with no additional abstraction layers on top (lowest level), which minimizes unnecessary overhead on a single node VM. Spark's results were skewed due to the first run, which required loading data and setting itself up. Subsequent Spark runs improved significantly due to in-memory caching and query optimisation, proving useful for iterative or repeated workloads. Unlike Spark, both Hive and MapReduce read directly from HDFS on every run without caching, which led to more consistent but comparatively slower execution times.

These results reflect the trade offs of each tool: Hive prioritises abstraction and convenience but takes longer due to its internal MapReduce translation (high overhead), Spark prioritises in-memory performance at scale but requires setup time and resources for the first run, and MapReduce prioritises simplicity and directness but does not have the optimisation and caching capabilities (that Spark has) needed for iterative or complex processes.

B6. Visualisation and interpretation

To understand the results obtained from the analytics pipeline better, visualizations were made to summarise both the system performance and the electricity consumption patterns.

B6.1 Tool Performance Comparison

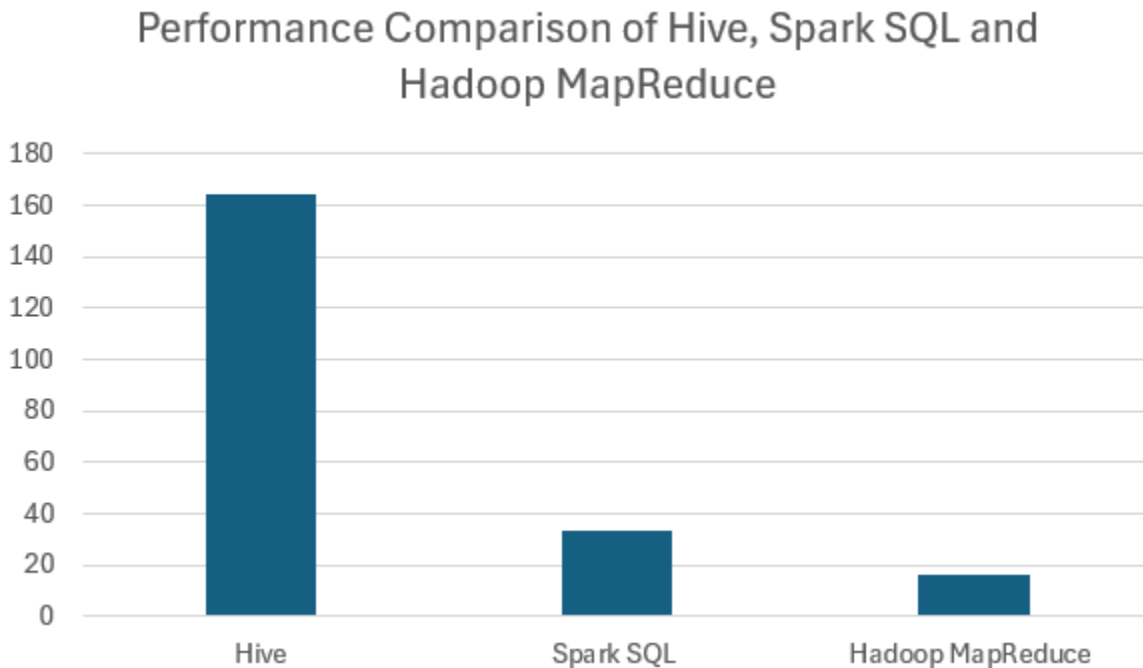


Figure B6.1 : Average execution time comparison of Hive, Spark SQL, and Hadoop MapReduce for the daily energy aggregation task.

The results show that Hive has the longest execution time (164.14 seconds) because Hive queries are translated into MapReduce jobs and involve more disk input/output operations. Spark SQL performs faster (33.73 seconds) due to its in-memory processing and query optimisation. Hadoop MapReduce shows the fastest execution time (16.36 seconds) in this experiment because the task was implemented directly using MapReduce framework without additional abstraction layers.

B6.2 Energy Consumption Insight

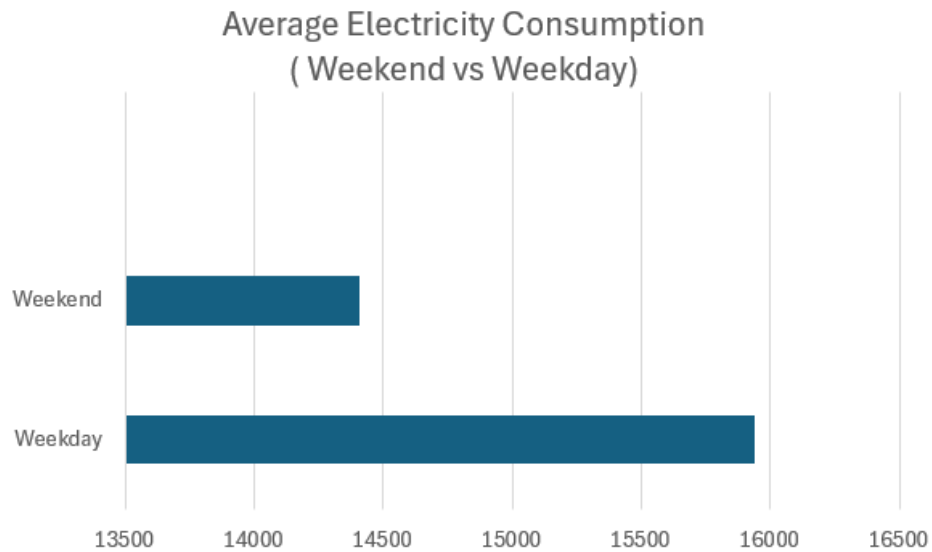


Figure B6.2 Average electricity consumption comparison between weekday and weekend.

The result shows that weekday electricity usage (approximately 15,936 MW) is higher than weekend usage (approximately 14,406 MW). This difference is likely due to increased commercial, industrial, and office activities during weekdays. Understanding this pattern helps energy providers better manage electricity demand and plan energy distribution more efficiently.

These visualizations not only show system performance and consumption patterns but also provide actionable intelligence for demand management and energy planning.

B7. Recommendations and Conclusion

Tool Suitability:

Spark SQL iterative analytics and feature engineering.

The average runtime of Spark (33.73 s) was significantly lower than Hive (164.14 s) and improved after the initial run due to caching and optimisation, making it well suited to repeated exploration, query refinement and multi-step feature workflow.

Java MapReduce to fixed large scale batch aggregations.

The MapReduce job had processed 121,273 hourly records onto 5,055 daily outputs, with the shortest average runtime (16.361 s) and consistent performance reliability over runs. It is thus suggested to be used for predictable types of daily rollups where throughput and stability are priorities.

Managed SQL reporting Hive, and with increased overhead.

Although Hive was the slowest in this comparison (164.14 s average), it remained useful in managing governed tables and reproducible reporting queries that were quite straightforward to maintain.

Scaling improvements

Data correctness checks: insert complete checks (e.g., predicted 24 hourly rows of each day per meter). The findings reveal that 2018-08-03 might be incomplete, and would influence total numbers on a daily basis and create false anomaly notifications.

Standardise timestamp parsing across tools: Datetime is defined as M/D/YYYY H:MM by the dataset, whereas DD/MM/YYYY H:MM is used by the MapReduce narrative. Apply a single parsing rule and normalise a DATE early to avoid unannounced shifts in dates.

Optimise storage layout: store cache curated results in Parquet/ORC, partition by year/month/day, and compress small files to minimise scan and metadata overhead at multi-meter scale.

Conclusion

The pipeline generated insights that could be used to make decisions, such as higher weekday than weekend demand (15,936.68 MW vs 14,405.98 MW) and an evening peak at approximately 20:00 (17,673.02 MW). Overall, Spark is recommended to be used in iterative analytics, MapReduce for stable large-scale batch rollups, and Hive for managed SQL reporting.

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Appendix

HIVE Query 2: Peak Hour Analysis (24-hour pattern)

This query analyzes the dataset by hour of the day to identify when electricity consumption peaks.

```
hive> SELECT ts as hour_of_day,
> AVG(aep_mw),2 as avg_load_mw,
> MAX(aep_mw),2 as max_load_mw
> FROM aep_hourly_parsed
> GROUP BY ts
> ORDER BY avg_load_mw DESC
> LIMIT 5;
```

hour_of_day	avg_load_mw	max_load_mw
2018-08-03 00:00:00	14809.0 2	14809.0 2
2018-08-02 23:00:00	15964.0 2	15964.0 2
2018-08-02 22:00:00	17001.0 2	17001.0 2
2018-08-02 21:00:00	17303.0 2	17303.0 2
2018-08-02 20:00:00	17673.0 2	17673.0 2

Time taken: 249.709 seconds, Fetched: 5 row(s)

Figure B2.9: HiveQL query execution and output identifying peak average load by hour.

The results indicate the highest average energy loads occur in the evening. Specifically, the peak average consumption happens at 20:00:00 (8:00 PM) with 17,673.02 MW, followed closely by 21:00:00 (9:00 PM) with 17,303.02 MW.

HIVE Query 3: Weekday vs Weekend Patterns

This query categorizes the data into weekdays and weekends to compare average daily energy requirements.

```
hive> SELECT
> CASE WHEN dayofweek(d) IN (1,7) THEN 'Weekend'
> ELSE 'Weekday' END as day_type,
> AVG(aep_mw),2 as avg_daily_mw
> FROM aep_hourly_parsed
> GROUP BY CASE WHEN dayofweek(d) IN (1,7) THEN 'Weekend' ELSE 'Weekday' END;
```

day_type	avg_daily_mw
Weekday	15936.68767385759 2
Weekend	14405.9855930246 2

Time taken: 65.287 seconds, Fetched: 2 row(s)

Figure B2.10: Query execution and output comparing weekday and weekend average loads

The output demonstrates a distinct operational pattern, showing that average electricity consumption is significantly higher on weekdays (15,936.68 MW) compared to weekends (14,405.98 MW).

SPARK SQL Query 2: Peak Hour Analysis (24-hour pattern)

Insight: At what hours of the day is electricity demand highest

```
scala> sqlContext.sql("""
  | SELECT
  |   hr,
  |   AVG(aep_mw) AS avg_load_mw,
  |   MAX(aep_mw) AS max_load_mw
  | FROM aep_hourly_parsed
  | GROUP BY hr
  | ORDER BY avg_load_mw DESC
  | LIMIT 5
  | """).show()
```

Time taken: 164.73 seconds

Rows fetched: 5

hr	avg_load_mw	max_load_mw
19	16868.728333992876	24538.0
20	16821.3351800554	24065.0
21	16763.806292045905	23822.0
18	16762.94004748714	24878.0
17	16542.038781163435	25140.0

Figure B5.1 Peak Hour Analysis

What the result shows:

The highest-demand hours are 19, 20, 21, 18, and 17, which means the evening period has the strongest energy usage. Hour 19 has the highest average load, while hour 17 has the highest recorded maximum load.

SPARK SQL Query 3: Weekday vs Weekend Patterns

Insight:

Does electricity usage differ between weekdays and weekends

```
scala> val result = sqlContext.sql("""
  | SELECT
  | CASE
  | WHEN dayofweek(d) IN (1,7) THEN 'Weekend'
  | ELSE 'Weekday'
  | END AS day_type,
  | ROUND(AVG(aep_mw),2) AS avg_energy_mw
  | FROM aep_hourly_parsed
  | GROUP BY
  | CASE
  | WHEN dayofweek(d) IN (1,7) THEN 'Weekend'
  | ELSE 'Weekday'
  | END
  | """).show()
```

Time taken: 147.47 seconds

Rows fetched: 2

day_type	avg_energy_mw
Weekend	14405.99
Weekday	15936.69

Figure B5.3: Weekday vs Weekend Energy Consumption

What the result shows:

The output shows:

- Weekend: 14405.99 MW
- Weekday: 15936.69 MW

This means weekday electricity usage is higher than weekend usage.

Report

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ASSESSMENT SUBMISSION DECLARATION FORM

Module Code:	ITS66904
Module Name:	BIG DATA TECHNOLOGIES
Assessment:	GROUP ASSIGNMENT (30%)

Student Name:	Rayhan Rizwan Manegar
Student ID:	0370148
Semester/Year:	Sem 5.5 Year 3

Dear Students,

Taylor's University/College upholds the highest standards of academic integrity. Students must complete their work honestly and ethically, ensuring that all submitted assessments reflect their own understanding and efforts. Any form of academic misconduct, including but not limited to plagiarism, collusion, contract cheating, falsification, and unauthorised use of Gen AI tools, is strictly prohibited and will be subject to disciplinary action in accordance with university regulations.

In this form, there are two (2) sections namely: **Section 1 Instruction on the use of Generative Artificial Intelligence** (to be completed by the module leader) and **Section 2 Student Acknowledgement and Declaration** (to be completed, signed and submitted by the students together with their assessment).

Students are responsible for properly acknowledging all sources, tools, and external contributions in their work. Misuse of AI-generated content, failure to attribute sources, or engagement in dishonest practices may result in penalties ranging from warnings to expulsion, as outlined in the **Student Handbook** (link).

This Assessment Submission Declaration Form is attached to the assessment brief. **Students are required to complete the Assessment Submission Declaration Form in any piece of work submitted** at Taylor's University/College, except for assessments completed and submitted in controlled environments. **For group assessments, each member of the group must complete an Assessment Submission Declaration Form.**

SECTION 1: INSTRUCTION ON THE USE OF GENERATIVE ARTIFICIAL INTELLIGENCE FOR THIS ASSESSMENT

The section below is to be completed by the module leader, i.e. module leader is to put a 'v' in the relevant checkbox.

☐ Guided use of Gen AI

Students are allowed to use Gen AI as stipulated by lecturers in the assessment brief:

1. Use lecturer-approved Gen AI tools only for certain parts of an assessment (e.g. brainstorming, proofreading etc.) or the entire assessment.
2. Use Gen AI for specific purposes only, such as generating drafts, refining language, or conducting exploratory analysis, but not for creating final outputs or core analytical tasks.

☒ Flexible use of Gen AI

You are allowed to use Gen AI tools in any way you find appropriate for the task, provided you acknowledge and explain their usage. This emphasises transparency, critical engagement, and ethical practices.

☐ Use of Gen AI is not applicable

Assessment is conducted under controlled environments (final exams, practical assessment without written assignment etc). Students are to complete the task independently, without the assistance of Gen AI tools.

Where Gen AI tools have been used, you must attach your chat history and an appendix listing the prompt used, the output generated by the generative AI tools, how the output was used and the pages where the AI-generated content can be found as per the table below:

AI Tool Used	Prompts	Outputs	How the output was changed for use into the assignment	Specify the page(s) in the assignment where the output is used
Example: ChatGPT	"Describe principles of marketing in 30 words"	The principles of marketing focus on understanding customer needs, creating value, building relationships, targeting specific markets, using effective promotion, ensuring product accessibility, pricing strategically, and fostering long-term loyalty and satisfaction.	The principles of marketing, as outlined by Kotler and Keller (2016), emphasize the importance of understanding customer needs, creating value, and building strong relationships. Key elements include targeting specific market segments, implementing effective promotional strategies, ensuring product accessibility, employing strategic pricing, and fostering long-term customer loyalty and satisfaction through a customer-centric approach (ChatGPT, 2024).	Page 1

The Gen AI-generated content must be appropriately cited and referenced. Any work submitted using AI tools without proper citation and referenced will be treated as though it was plagiarised.

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- Counseling or cautioning.
- Awarding zero marks for the assignment.
- Failing the module.
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If an assignment is suspected of improper AI use or other academic integrity violations, the university will follow the official Academic Integrity Investigation Process, which includes:

- Initial Assessment**-The lecturer assesses inconsistencies in writing style, AI detection reports, or academic dishonesty concerns.
- Filing of Report** -A Report on Academic Misconduct (THE-ACA-FORM-AINT) is submitted to the Academic Integrity Officer (AIO).
- Review & Evidence Collection** – The AIO compiles relevant materials, including previous submissions, Turnitin reports, and AI usage logs.
- Student Explanation** – The student may be required to provide additional evidence of their work process, including drafts and source materials.
- Disciplinary Review** – The Student Disciplinary Committee (SDC) will determine penalties if misconduct is confirmed
- Final Decision & Notification** – The student receives formal notification of the decision, and records are updated accordingly.

For full details, refer to the Student Handbook ([link](#))

SECTION 2: STUDENT ACKNOWLEDGEMENT AND DECLARATION

(*This section is to be completed by students. Students must submit the completed and signed copy of this form together with their assessment)

Acknowledgment by the student

When Gen AI is allowed to be used, acknowledgement of how Gen AI is used can be included as follows: (Place a tick '✓' in the checkbox where relevant)

	Type	Acknowledgement
✓	Gen AI is allowed but student chooses not to use it	I did not use any Gen AI tools for this assignment.
	Generate content which is modified in the final assessment submission	I acknowledge the use of to general materials such as in this assessment. I have also provided the prompt used, the output generated by the Gen AI tools, how the output was used and the pages where the Gen AI-generated content can be found.
	Use Gen AI to polish language before further modification for final submission	I acknowledge the use of to improve the academic tone and accuracy of language, including grammatical structures, punctuation and vocabulary. I have also provided the prompt used, the output generated by the Gen AI tools, how the output was modified further to better represent my tone and style of writing. <i>Note: Please modify the above as per your lecturers' needs</i>
	I attached Appendix in this assignment the chat history, including all the prompts that I used and the output from the Gen AI tool(s) for this assignment.	

The use of GenAI has to be referenced appropriately according to the referencing style as required:

Referencing Style*	In-text citation	Reference
APA	... (AI platform, year)	Company. (year). <i>Platform Used</i> (version) [model]. URL Example: OpenAI. (2023). <i>ChatGTP</i> (Mar 14 version) [Large language model]. https://chat.openai.com/chat
MLA	 ("prompt").	"Description of prompt" prompt. <i>Name of AI tool</i> , version of AI tool, Company, date text was generated. URL. Example (Text): "Identify the themes in Mcteague by Frank Norris" prompt. ChatGPT, 21 November version, OpenAI, 4 Dec. 2023. https://chat.openai.com/share/2f2be19d-eadd-4151-8ceb-0785319074b3

		Example (Image): Fig. 2. "Create impressionist painting of a cat using neutral colors" prompt. DALL-E, version 2, OpenAI, 27 Nov. 2023. https://labs.openai.com/
Harvard	... (AI platform year)	There is no guideline for Harvard style on referencing the use of AI. The following format is recommended by most universities. Name of AI (Year of communication) Medium of communication. Receiver of communication, Day/month of communication. Example OpenAI ChatGPT (2023) ChatGPT response to Mary Jane, 25 January 2024

*Please refer to the latest referencing style.

Declaration

I confirm that:

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- I confirm that my work or any part of this assessment has neither been previously and is not concurrently submitted for any other programme at Taylor's University or any other institution, save except when re-use of the same work is permitted by the module leader.
- I acknowledge that using Gen AI or any external assistance without proper attribution constitutes academic misconduct and may be sanctioned accordingly.
- I understand that if there are indications of academic integrity breaches, including improper Gen AI use, my work will be subject to investigation.
- This assignment is my own work, and I have properly acknowledged all sources, tools, and external contributions, including the use of Gen AI where applicable.
- I acknowledge and authorize the submission and/or storage of my work in a database for the purpose of verifying its originality and/or conducting tests using artificial intelligence software, and I hereby consent to this process.
- I acknowledge that this submission is subject to Taylor's University/College Academic Integrity Procedure (THE-ACA-PROC-AINT) and all applicable university regulations.

Name & Student ID	Rayhan Rizwan 0370148
Signature	
Date	02/03/2026



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ASSESSMENT SUBMISSION DECLARATION FORM

Module Code:	ITS66904
Module Name:	BIG DATA TECHNOLOGIES
Assessment:	GROUP ASSIGNMENT (30%)

Student Name:	Melanie Grace Yee Fone Ip Wan Fat
Student ID:	0373904
Semester/Year:	Sem 4.5 Year 2

Dear Students,

Taylor's University/College upholds the highest standards of academic integrity. Students must complete their work honestly and ethically, ensuring that all submitted assessments reflect their own understanding and efforts. Any form of academic misconduct, including but not limited to plagiarism, collusion, contract cheating, falsification, and unauthorised use of Gen AI tools, is strictly prohibited and will be subject to disciplinary action in accordance with university regulations.

In this form, there are two (2) sections namely: **Section 1 Instruction on the use of Generative Artificial Intelligence** (to be completed by the module leader) and **Section 2 Student Acknowledgement and Declaration** (to be completed, signed and submitted by the students together with their assessment).

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SECTION 1: INSTRUCTION ON THE USE OF GENERATIVE ARTIFICIAL INTELLIGENCE FOR THIS ASSESSMENT

The section below is to be completed by the module leader, i.e. module leader is to put a 'v' in the relevant checkbox.

☐ Guided use of Gen AI

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☐ Use of Gen AI is not applicable

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AI Tool Used	Prompts	Outputs	How the output was changed for use into the assignment	Specify the page(s) in the assignment where the output is used
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INFORMATION ON PENALTIES RELATING TO ACADEMIC MISCONDUCT

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The penalties that can be imposed on a student who is found to have engaged in academic misconduct include, but not limited to:

- Counseling or cautioning.
- Awarding zero marks for the assignment.
- Failing the module.
- Suspension from enrollment at Taylor's University/College.
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Acknowledgment by the student

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	Generate content which is modified in the final assessment submission	I acknowledge the use of to general materials such as in this assessment. I have also provided the prompt used, the output generated by the Gen AI tools, how the output was used and the pages where the Gen AI-generated content can be found.
	Use Gen AI to polish language before further modification for final submission	I acknowledge the use of to improve the academic tone and accuracy of language, including grammatical structures, punctuation and vocabulary. I have also provided the prompt used, the output generated by the Gen AI tools, how the output was modified further to better represent my tone and style of writing. <i>Note: Please modify the above as per your lecturers' needs</i>
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Declaration

I confirm that:

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- I confirm that my work or any part of this assessment has neither been previously and is not concurrently submitted for any other programme at Taylor's University or any other institution, save except when re-use of the same work is permitted by the module leader.
- I acknowledge that using Gen AI or any external assistance without proper attribution constitutes academic misconduct and may be sanctioned accordingly.
- I understand that if there are indications of academic integrity breaches, including improper Gen AI use, my work will be subject to investigation.
- This assignment is my own work, and I have properly acknowledged all sources, tools, and external contributions, including the use of Gen AI where applicable.
- I acknowledge and authorize the submission and/or storage of my work in a database for the purpose of verifying its originality and/or conducting tests using artificial intelligence software, and I hereby consent to this process.
- I acknowledge that this submission is subject to Taylor's University/College Academic Integrity Procedure (THE-ACA-PROC-AINT) and all applicable university regulations.

Name & Student ID	Melanie Grace Yee Fone Ip Wan Fat 0373904
Signature	
Date	02/03/2026



Taylor's University / Taylor's College

ASSESSMENT SUBMISSION DECLARATION FORM

Module Code:	ITS66904
Module Name:	Big Data Technologies
Assessment:	Group Assignment

Student Name:	Leroy Chiew Ruo-Wey
Student ID:	0370618
Semester/Year:	Sem 3.5 Year 2

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	Type	Acknowledgement
	Gen AI is allowed but student chooses not to use it	I did not use any Gen AI tools for this assignment.
	Generate content which is modified in the final assessment submission	I acknowledge the use of to general materials such as ...demo cognitive computing application.... in this assessment. I have also provided the prompt used, the output generated by the Gen AI tools, how the output was used and the pages where the Gen AI-generated content can be found.
✓	Use Gen AI to polish language before further modification for final submission	I acknowledge the use of Grammarly..... to improve the academic tone and accuracy of language, including grammatical structures, punctuation and vocabulary. I have also provided the prompt used, the output generated by the Gen AI tools, how the output was modified further to better represent my tone and style of writing. <i>Note: Please modify the above as per your lecturers' needs</i>
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Name & Student ID	Leroy Chiew Ruo-Wey 0370618
Signature	
Date	10/03/2026



Taylor's University / Taylor's College

ASSESSMENT SUBMISSION DECLARATION FORM

Module Code:	ITS66904
Module Name:	Big Data Technologies
Assessment:	Assessment 02

Student Name:	Lai Ze Qi
Student ID:	0371936
Semester/Year:	Sem 3.5 Year 2

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Name & Student ID	Lai Ze Qi 0371936
Signature	
Date	10/03/2026



Taylor's University / Taylor's College

ASSESSMENT SUBMISSION DECLARATION FORM

Module Code:	ITS66904
Module Name:	BIG DATA TECHNOLOGIES
Assessment:	GROUP ASSIGNMENT (30%)

Student Name:	Lai Ze Xuan
Student ID:	0371938
Semester/Year:	Sem 3.5 Year 2

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- Filing of Report** -A Report on Academic Misconduct (THE-ACA-FORM-AINT) is submitted to the Academic Integrity Officer (AIO).
- Review & Evidence Collection** – The AIO compiles relevant materials, including previous submissions, Turnitin reports, and AI usage logs.
- Student Explanation** – The student may be required to provide additional evidence of their work process, including drafts and source materials.
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For full details, refer to the Student Handbook ([link](#))

SECTION 2: STUDENT ACKNOWLEDGEMENT AND DECLARATION

(*This section is to be completed by students. Students must submit the completed and signed copy of this form together with their assessment)

Acknowledgment by the student

When Gen AI is allowed to be used, acknowledgement of how Gen AI is used can be included as follows: (Place a tick '✓' in the checkbox where relevant)

	Type	Acknowledgement
✓	Gen AI is allowed but student chooses not to use it	I did not use any Gen AI tools for this assignment.
	Generate content which is modified in the final assessment submission	I acknowledge the use of to general materials such as in this assessment. I have also provided the prompt used, the output generated by the Gen AI tools, how the output was used and the pages where the Gen AI-generated content can be found.
	Use Gen AI to polish language before further modification for final submission	I acknowledge the use of to improve the academic tone and accuracy of language, including grammatical structures, punctuation and vocabulary. I have also provided the prompt used, the output generated by the Gen AI tools, how the output was modified further to better represent my tone and style of writing. <i>Note: Please modify the above as per your lecturers' needs</i>
	I attached Appendix in this assignment the chat history, including all the prompts that I used and the output from the Gen AI tool(s) for this assignment.	

The use of GenAI has to be referenced appropriately according to the referencing style as required:

Referencing Style*	In-text citation	Reference
APA	... (AI platform, year)	Company. (year). <i>Platform Used</i> (version) [model]. URL Example: OpenAI. (2023). <i>ChatGTP</i> (Mar 14 version) [Large language model]. https://chat.openai.com/chat
MLA	 ("prompt").	"Description of prompt" prompt. <i>Name of AI tool</i> , version of AI tool, Company, date text was generated. URL. Example (Text): "Identify the themes in Mcteague by Frank Norris" prompt. ChatGPT, 21 November version, OpenAI, 4 Dec. 2023. https://chat.openai.com/share/2f2be19d-eadd-4151-8ceb-0785319074b3

		Example (Image): Fig. 2. "Create impressionist painting of a cat using neutral colors" prompt. DALL-E, version 2, OpenAI, 27 Nov. 2023. https://labs.openai.com/
Harvard	... (AI platform year)	There is no guideline for Harvard style on referencing the use of AI. The following format is recommended by most universities. Name of AI (Year of communication) Medium of communication. Receiver of communication, Day/month of communication. Example OpenAI ChatGPT (2023) ChatGPT response to Mary Jane, 25 January 2024

*Please refer to the latest referencing style.

Declaration

I confirm that:

- I understand what constitutes academic integrity violations, including plagiarism, collusion, fabrication, falsification, contract cheating, and improper use of Gen AI.
- I confirm that my work or any part of this assessment has neither been previously and is not concurrently submitted for any other programme at Taylor's University or any other institution, save except when re-use of the same work is permitted by the module leader.
- I acknowledge that using Gen AI or any external assistance without proper attribution constitutes academic misconduct and may be sanctioned accordingly.
- I understand that if there are indications of academic integrity breaches, including improper Gen AI use, my work will be subject to investigation.
- This assignment is my own work, and I have properly acknowledged all sources, tools, and external contributions, including the use of Gen AI where applicable.
- I acknowledge and authorize the submission and/or storage of my work in a database for the purpose of verifying its originality and/or conducting tests using artificial intelligence software, and I hereby consent to this process.
- I acknowledge that this submission is subject to Taylor's University/College Academic Integrity Procedure (THE-ACA-PROC-AINT) and all applicable university regulations.

Name & Student ID	Lai Ze Xuan 0371938
Signature	
Date	02/03/2026



Taylor's University / Taylor's College

ASSESSMENT SUBMISSION DECLARATION FORM

Module Code:	ITS66904
Module Name:	BIG DATA TECHNOLOGIES
Assessment:	GROUP ASSIGNMENT (30%)

Student Name:	Hammad Ruxshan
Student ID:	0356865
Semester/Year:	Sem 5.5 Year 3

Dear Students,

Taylor's University/College upholds the highest standards of academic integrity. Students must complete their work honestly and ethically, ensuring that all submitted assessments reflect their own understanding and efforts. Any form of academic misconduct, including but not limited to plagiarism, collusion, contract cheating, falsification, and unauthorised use of Gen AI tools, is strictly prohibited and will be subject to disciplinary action in accordance with university regulations.

In this form, there are two (2) sections namely: **Section 1 Instruction on the use of Generative Artificial Intelligence** (to be completed by the module leader) and **Section 2 Student Acknowledgement and Declaration** (to be completed, signed and submitted by the students together with their assessment).

Students are responsible for properly acknowledging all sources, tools, and external contributions in their work. Misuse of AI-generated content, failure to attribute sources, or engagement in dishonest practices may result in penalties ranging from warnings to expulsion, as outlined in the **Student Handbook** (link).

This Assessment Submission Declaration Form is attached to the assessment brief. **Students are required to complete the Assessment Submission Declaration Form in any piece of work submitted** at Taylor's University/College, except for assessments completed and submitted in controlled environments. **For group assessments, each member of the group must complete an Assessment Submission Declaration Form.**

SECTION 1: INSTRUCTION ON THE USE OF GENERATIVE ARTIFICIAL INTELLIGENCE FOR THIS ASSESSMENT

The section below is to be completed by the module leader, i.e. module leader is to put a 'v' in the relevant checkbox.

☐ Guided use of Gen AI

Students are allowed to use Gen AI as stipulated by lecturers in the assessment brief:

1. Use lecturer-approved Gen AI tools only for certain parts of an assessment (e.g. brainstorming, proofreading etc.) or the entire assessment.
2. Use Gen AI for specific purposes only, such as generating drafts, refining language, or conducting exploratory analysis, but not for creating final outputs or core analytical tasks.

☒ Flexible use of Gen AI

You are allowed to use Gen AI tools in any way you find appropriate for the task, provided you acknowledge and explain their usage. This emphasises transparency, critical engagement, and ethical practices.

☐ Use of Gen AI is not applicable

Assessment is conducted under controlled environments (final exams, practical assessment without written assignment etc). Students are to complete the task independently, without the assistance of Gen AI tools.

Where Gen AI tools have been used, you must attach your chat history and an appendix listing the prompt used, the output generated by the generative AI tools, how the output was used and the pages where the AI-generated content can be found as per the table below:

AI Tool Used	Prompts	Outputs	How the output was changed for use into the assignment	Specify the page(s) in the assignment where the output is used
Example: ChatGPT	"Describe principles of marketing in 30 words"	The principles of marketing focus on understanding customer needs, creating value, building relationships, targeting specific markets, using effective promotion, ensuring product accessibility, pricing strategically, and fostering long-term loyalty and satisfaction.	The principles of marketing, as outlined by Kotler and Keller (2016), emphasize the importance of understanding customer needs, creating value, and building strong relationships. Key elements include targeting specific market segments, implementing effective promotional strategies, ensuring product accessibility, employing strategic pricing, and fostering long-term customer loyalty and satisfaction through a customer-centric approach (ChatGPT, 2024).	Page 1

The Gen AI-generated content must be appropriately cited and referenced. Any work submitted using AI tools without proper citation and referenced will be treated as though it was plagiarised.

INFORMATION ON PENALTIES RELATING TO ACADEMIC MISCONDUCT

Any student found to have engaged in academic misconduct, including but not limited to plagiarism, falsification, contract cheating, or improper AI usage, will be subject to disciplinary action. Penalties may include but are not limited to plagiarism, falsification, contract cheating, or improper AI usage, and will be subject to disciplinary action. Penalties may include but are not limited to:

The penalties that can be imposed on a student who is found to have engaged in academic misconduct include, but not limited to:

- Counseling or cautioning.
- Awarding zero marks for the assignment.
- Failing the module.
- Suspension from enrollment at Taylor's University/College.
- Expulsion from Taylor's University/College.

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Name & Student ID	Hammad Ruxshan 0356865
Signature	
Date	02/03/2026